

All-pair coalescence scan

$s = 0.002$ | saved simulations, newly decoded with Gamma-SMC

2% introgression pulse + selection starting at 50 kya

100 selected regions + 1,000 neutral regions

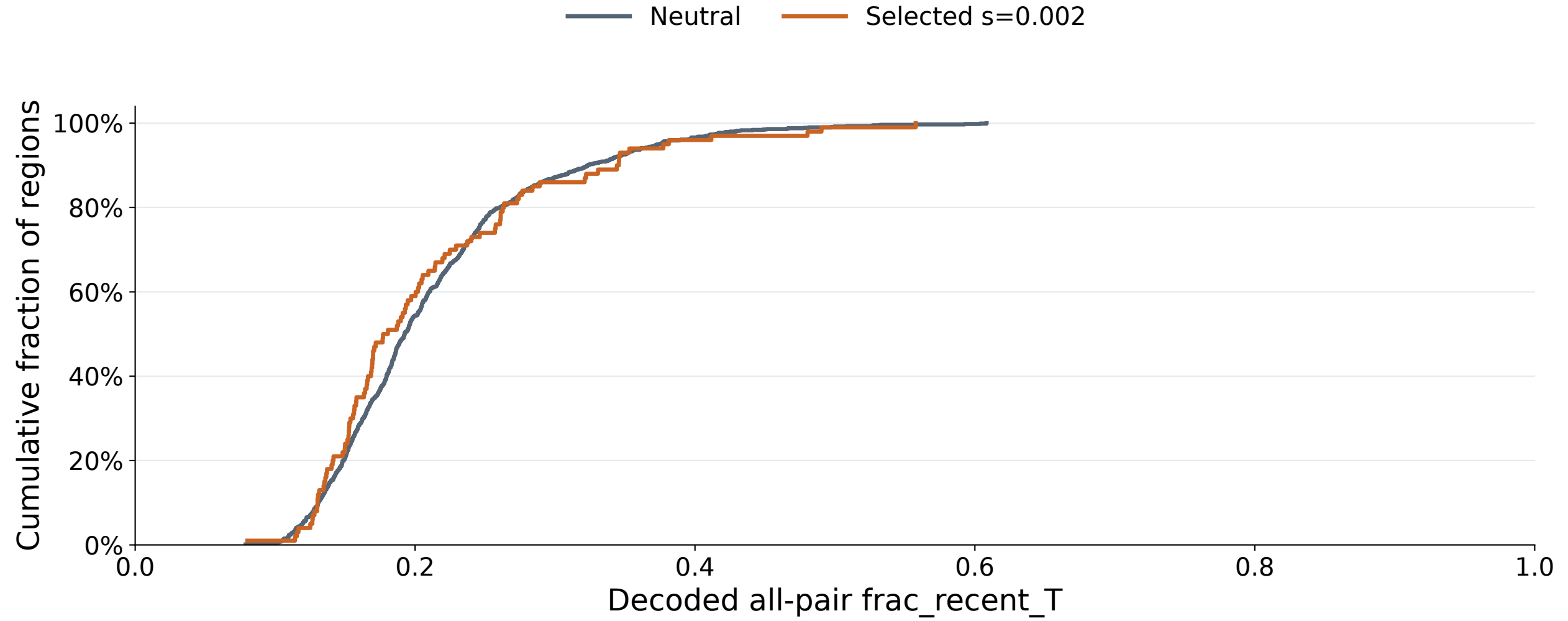
400 haplotypes; 10,000 fixed sampled pairs; isolated 10 Mb

$$S_T = \frac{n_{\text{pairs with decoded TMRCA} < T}}{10,000}$$

Posterior-mean TMRCA hard calls | fixed-coordinate neutral calibration

The all-pair score at the fixed focal coordinate

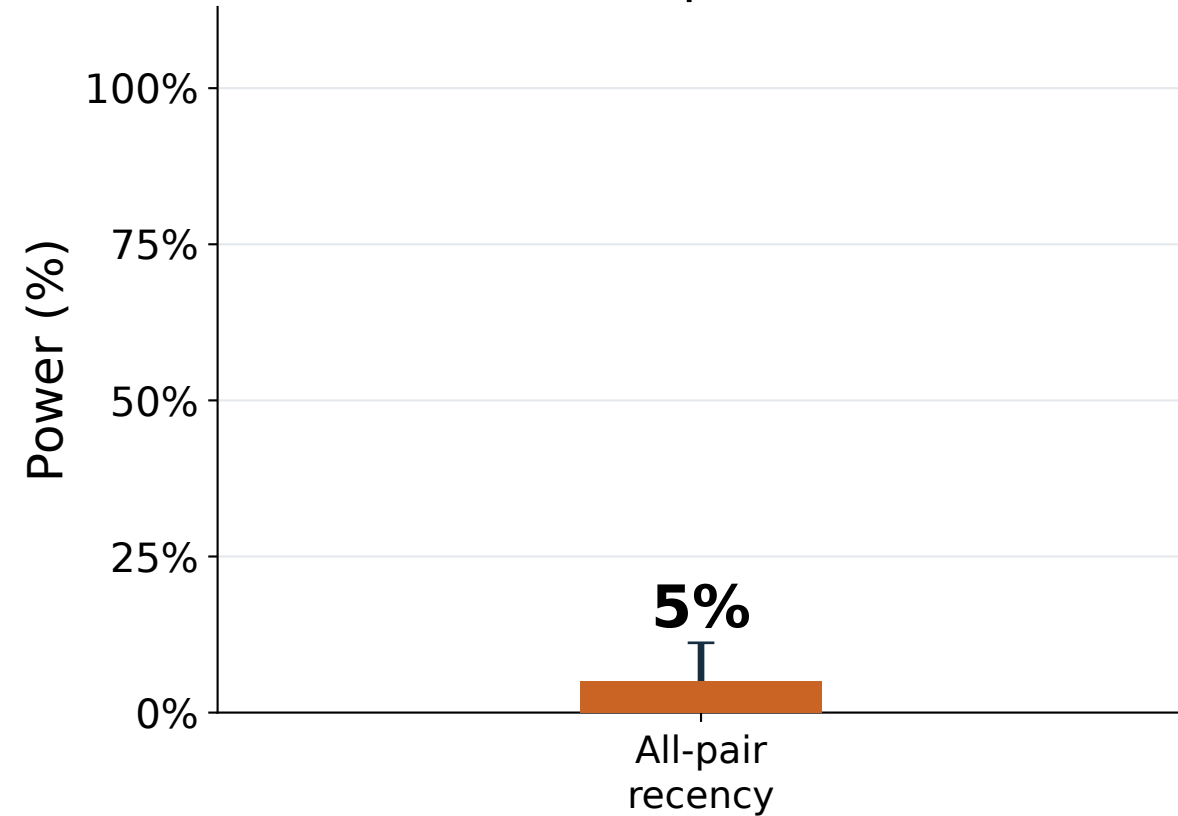
Gamma-SMC | T = 50 kya | 100 selected versus 1,000 neutral regions



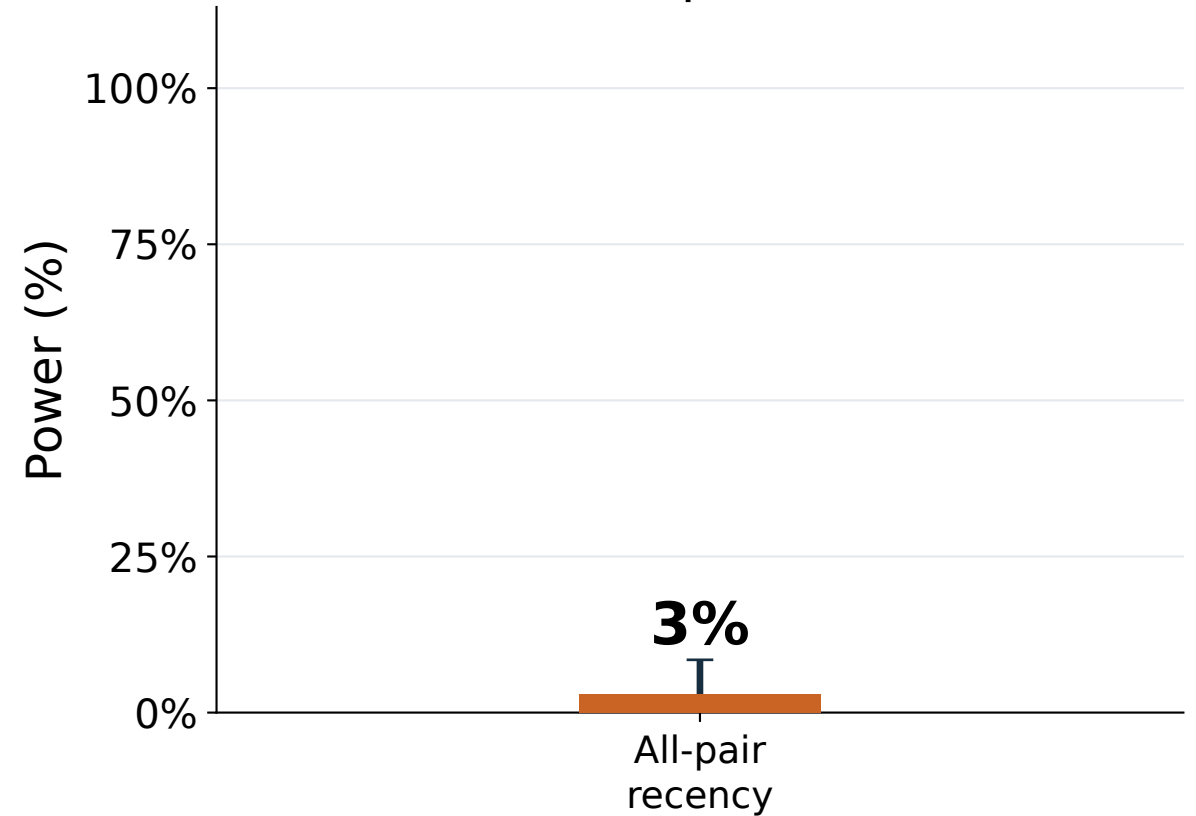
Decoded all-pair power at $s = 0.002$

T = 50 kya | 100 selected replicates | error bars: 95% Wilson intervals

Nominal $p \leq 0.05$

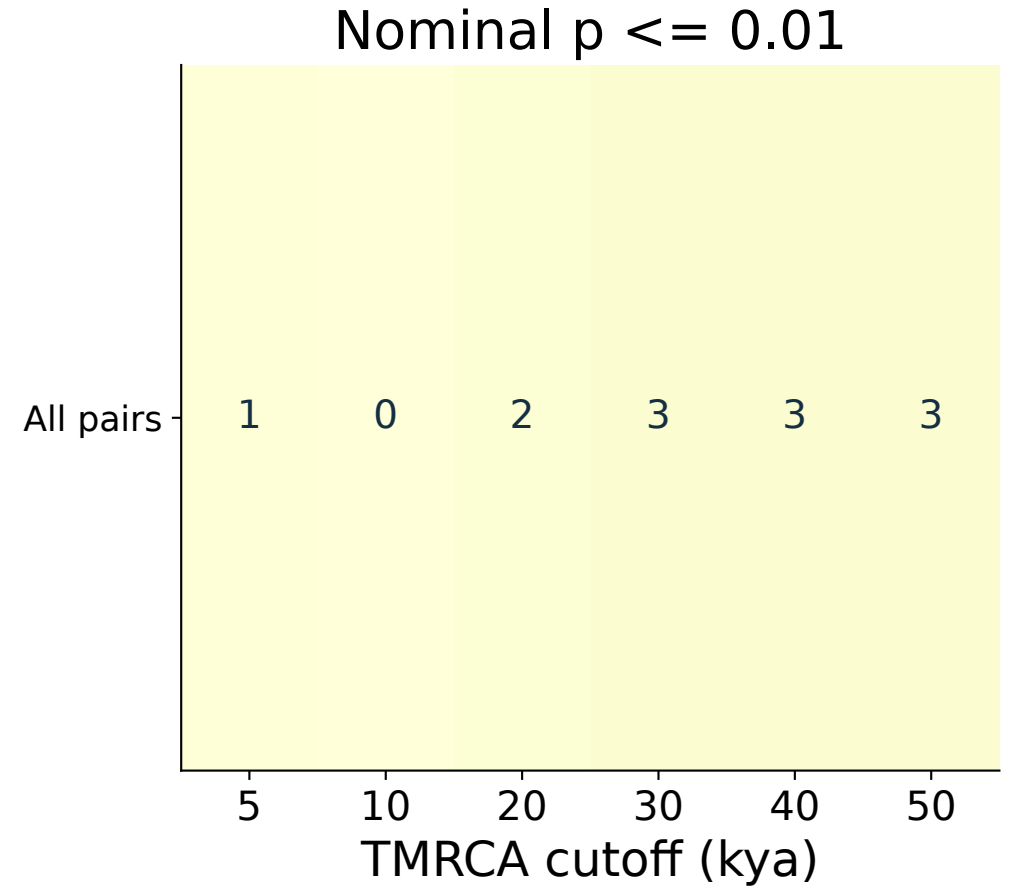
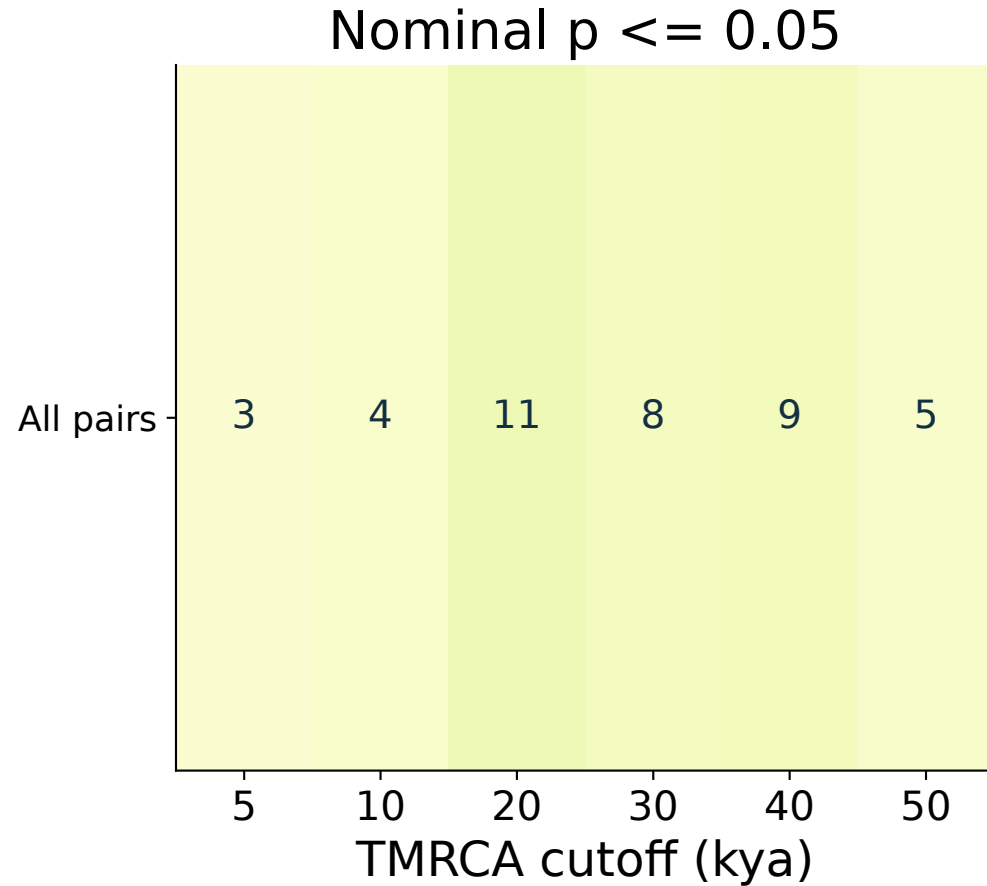


Nominal $p \leq 0.01$



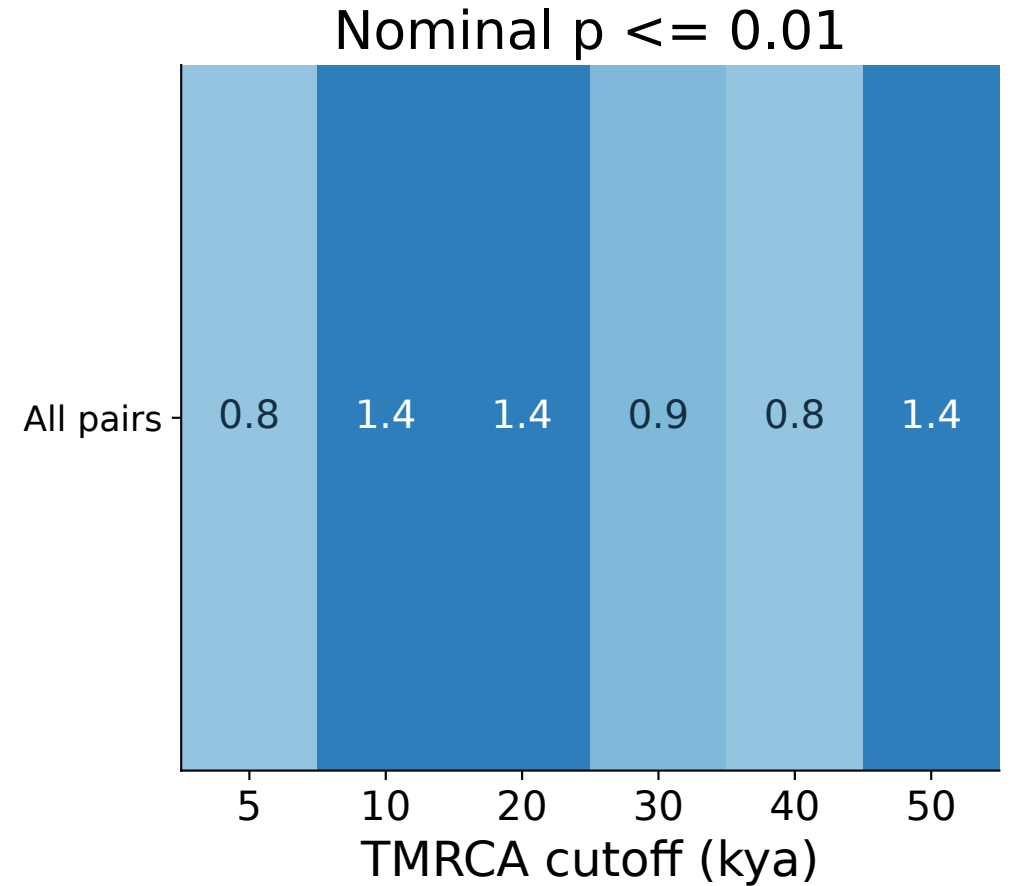
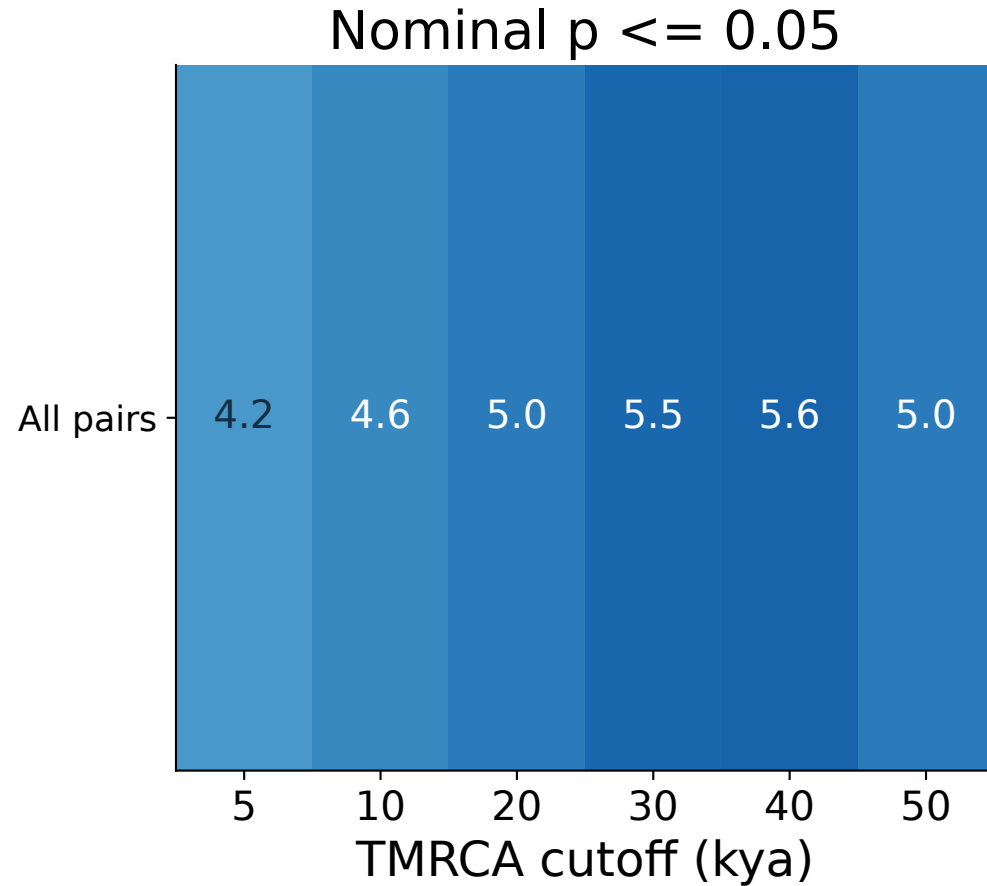
TMRCA cutoffs change decoded power

$s = 0.002$ | 100 selected replicates per cell | percentages



Decoded neutral false-positive rates

$s = 0.002$ | 1,000 neutral positions

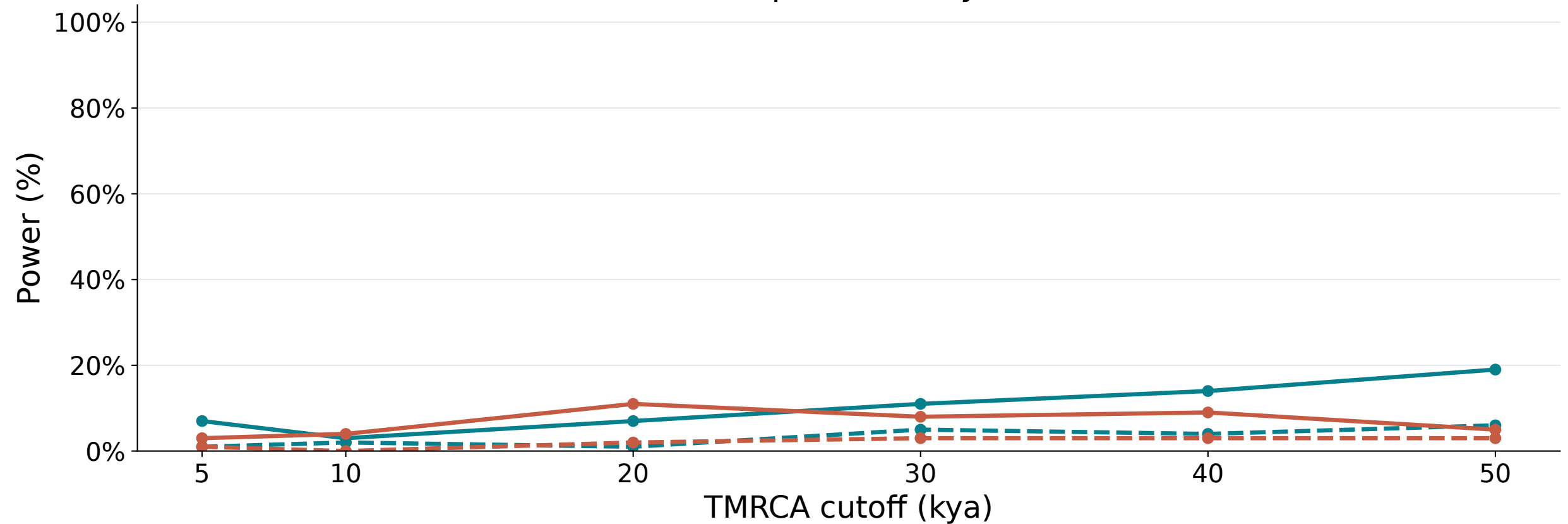


Validation: true versus decoded TMRCA

$s = 0.002$ | same saved genealogies, selected allele, and 10,000 pairs

—●— Truth, $p \leq 0.05$ -●- Truth, $p \leq 0.01$ —●— Decoded, $p \leq 0.05$ -●- Decoded, $p \leq 0.01$

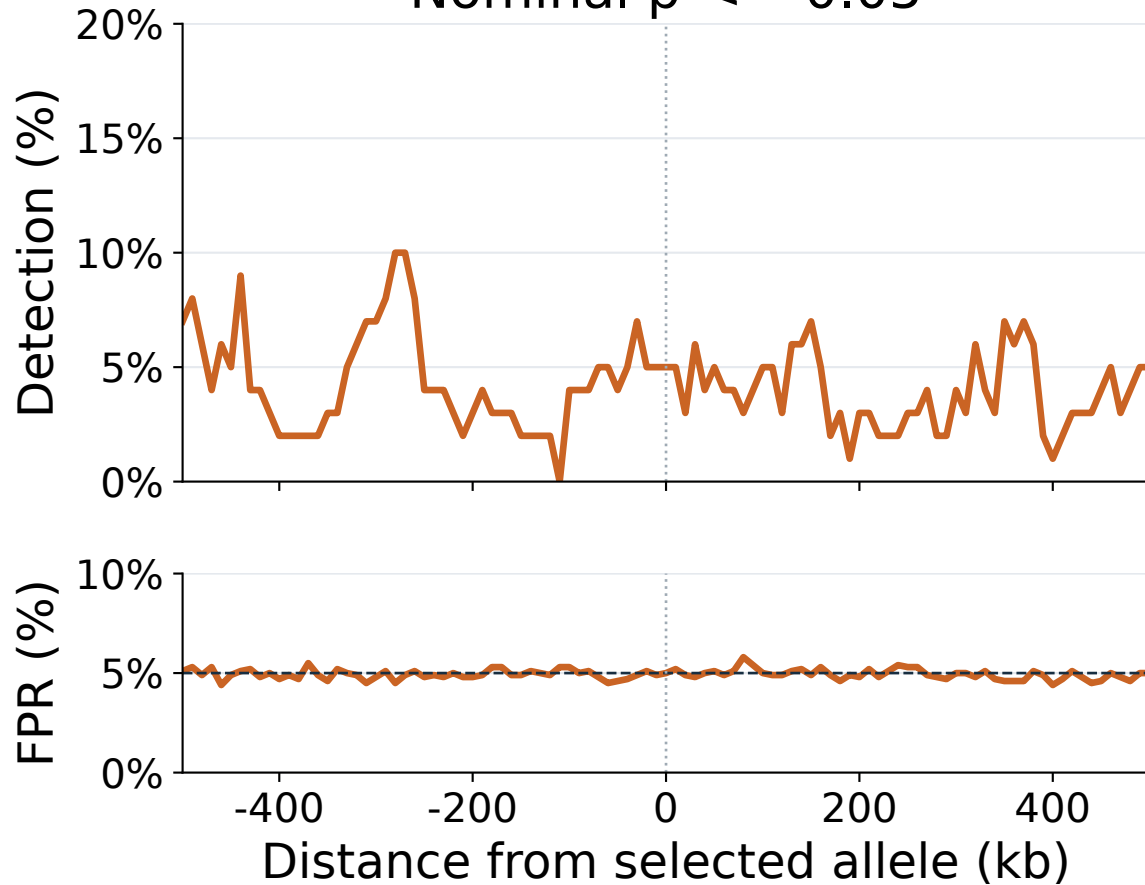
All-pair recency



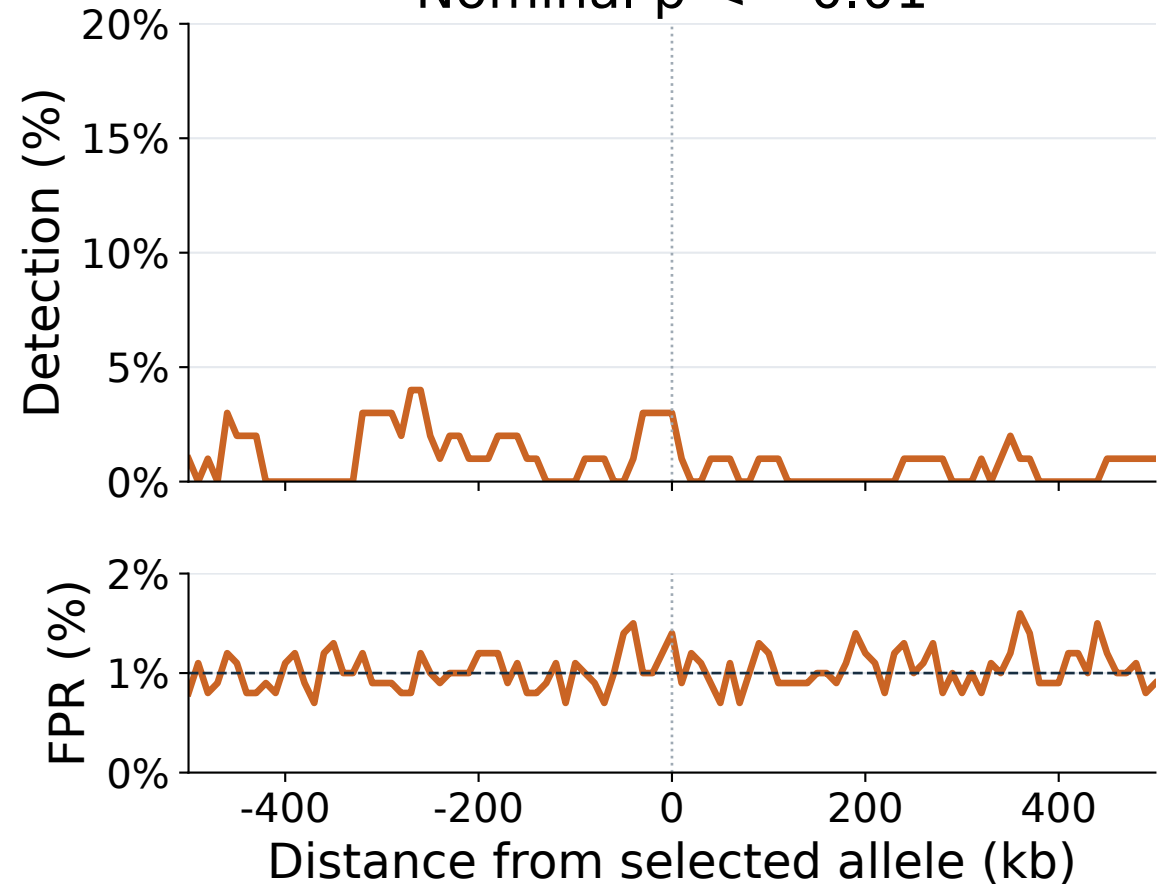
Spatial extent of the decoded all-pair signal

$s = 0.002$ | $T = 50$ kya | each stride uses its own matched neutral coordinate

Nominal $p \leq 0.05$

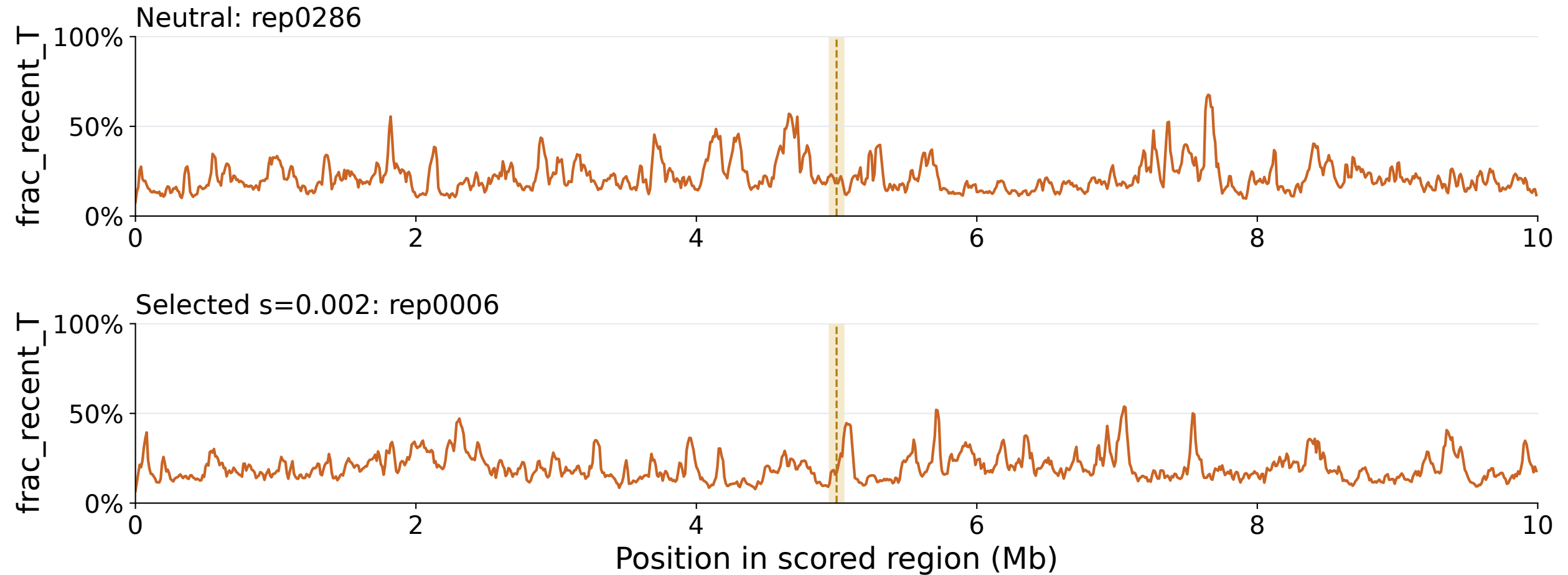


Nominal $p \leq 0.01$



Example decoded all-pair profiles

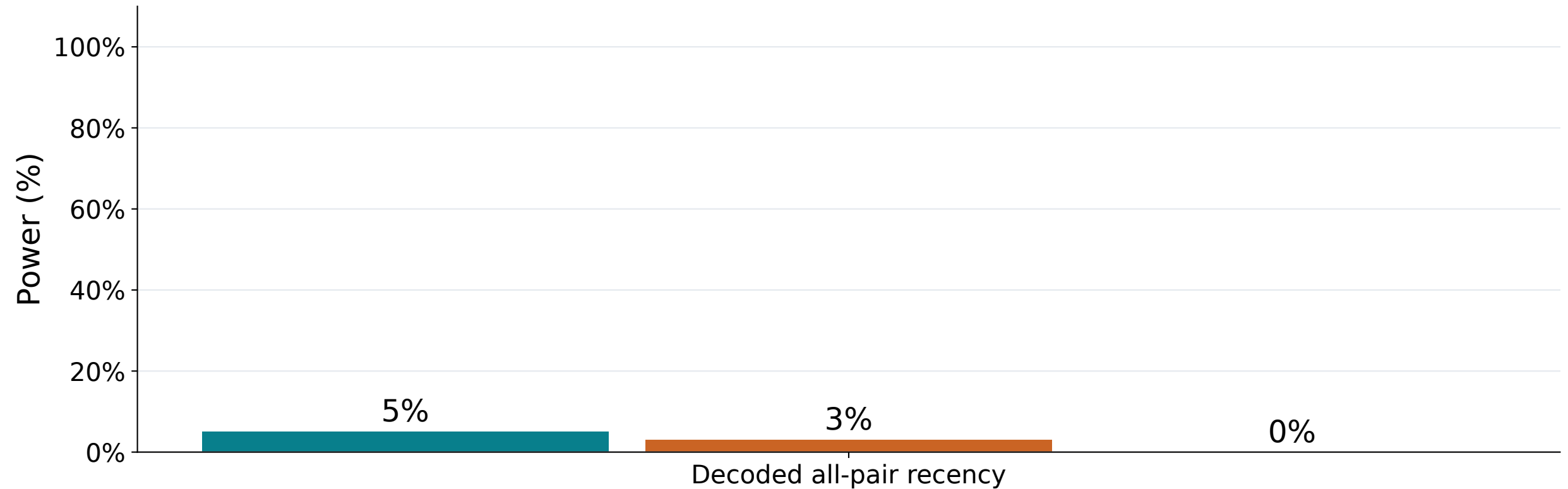
Examples chosen by median scores or allele frequency, independently of significance



Power at stricter positional thresholds

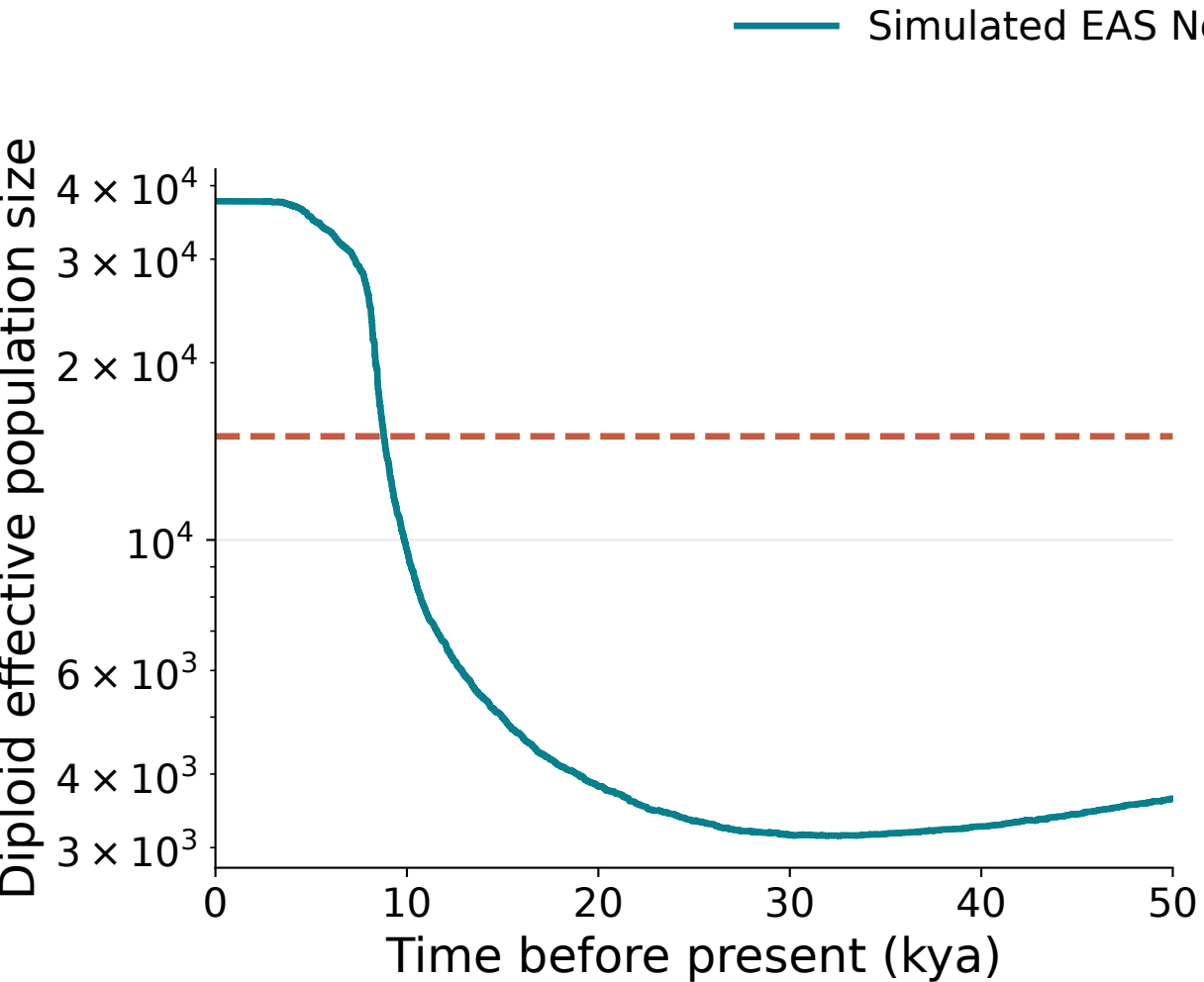
s = 0.002 | decoded TMRCA < 50 kya | p=0.001 uses pooled ranks

■ p ≤ 0.05 (5-fold) ■ p ≤ 0.01 (5-fold) ■ p ≤ 0.001 (pooled)



Demography is shared by neutral and selected simulations

The generative EAS history varies with time; the current decoder uses a fixed Ne



Archaic branch

Split: 700 kya

Baseline Ne: 3,600

Pulse bottleneck: Ne=10
for 100 generations

Mutation rate: 1.25×10^{-8}

Recombination rate: 1×10^{-8}

Generation time: 25 years